know, shards are Lucene instances that hold the physical data that's logically associated with an index.

When we index a document, the engine decides which shard the document will be housed in based on the routing algorithm (discussed in section 3.5). Each shard comes with heap memory, and when a document is indexed, the document is first pushed into the shard's in-memory buffer. The document is held in this in-memory buffer until a refresh occurs. Lucene's scheduler issues a refresh every second to collect all the documents available in the in-memory buffer, and then it creates a new segment with these documents. The segment consists of the document data and inverted indexes. Data is first written to the filesystem cache and then committed to the physical disk.

Because I/O operations are expensive, Lucene avoids frequent I/O operations when writing data to the disk. Hence, it waits for the refresh interval (one second), after which the documents are bundled to be pushed into segments. Once the documents are moved to segments, they are made available for searching.

Apache Lucene is an intelligent library when dealing with data writes and reads. After pushing the documents to a new segment (during the refresh operation), it waits until three segments are formed. It uses a three-segment-merge pattern to merge the segments to create new segments: whenever three segments are ready, Lucene instantiates a new one by merging them. Then it waits for three more segments to be created so it can create a new one, and so on. Every three segments merge to create another segment, as shown in figure 5.5.

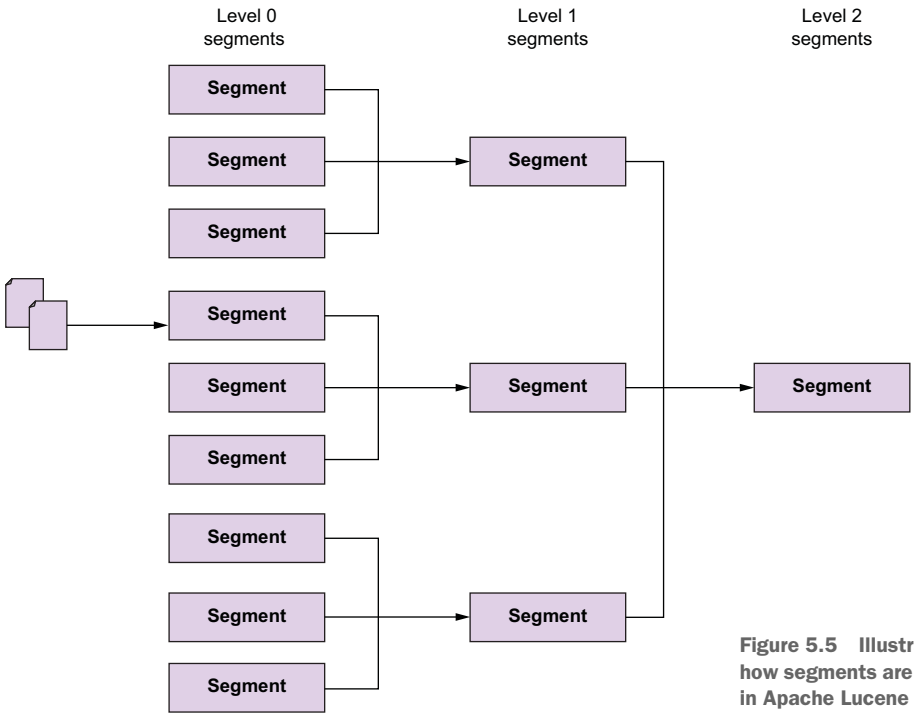


Figure 5.5 Illustration of how segments are merged in Apache Lucene